

Lab 1 Results Summary

Deep Generative Genre Remastering — Deconstruction Encoder

Demo-focused compact report

Objective

Lab 1 establishes the deconstruction encoder used throughout the project. The goal is to separate short music clips into:

- z_{content} : melody/rhythm/harmonic structure that should remain style-invariant
- z_{style} : genre and timbral cues that should remain style-informative
- **music gate**: a binary head that distinguishes music from non-music

The run summarized here is:

`saves/lab1_run_combo_af_gate_exit_v2/latest.pt`

What We Measured

Metric	Meaning	Goal
Style probe accuracy	Can a probe recover genre/source from z_{style} ?	High
Content leakage above baseline	How much style information still leaks through z_{content} above chance?	Low
Gate ROC AUC	How well does the music gate separate music from speech/non-music?	High

Observed Results

Metric	Observed	Threshold	Status
Style probe accuracy	0.9417	≥ 0.85	Pass
Content leakage above baseline	0.1083	≤ 0.15	Pass
Gate ROC AUC	0.9299	≥ 0.90	Pass

Additional gate details from the same audit:

- Gate accuracy at threshold 0.5: 0.8602
- False positive rate at threshold 0.5: 0.1361
- Music recall at threshold 0.5: 0.8551
- When recalibrated to roughly 2% FPR, recall drops to 0.5579

Interpretation

- **z_{style} is working:** the style probe reaches 94.17% accuracy, which means genre/source information is strongly present in the style branch.
- **z_{content} is meaningfully disentangled:** leakage is only 0.1083 above the 0.5 chance baseline, so style information is substantially suppressed in the content branch.
- **The music gate separates classes well overall:** ROC AUC is 0.9299, which is strong enough for the demo story and downstream filtering.
- **The remaining weakness is calibration, not representation:** the gate passes the AUC threshold, but a strict low-FPR operating point still sacrifices recall. That means Lab 1 succeeds as a feature-learning stage even though the binary gate threshold can still be tuned later.

Why Lab 1 Matters for the Rest of the Pipeline

These results justify using Lab 1 outputs in later stages:

- Lab 2 uses the style-bearing side to build a target genre space.
- Lab 3 uses the content side as a melody-preservation anchor during remastering.
- Lab 4 relies on the same structural representation to preserve continuity over long-form generation.

Demo Takeaway

For the demo, the key result is simple: **Lab 1 successfully produced a usable musical skeleton.** The content representation is sufficiently style-suppressed, the style representation is strongly discriminative, and the gate is reliable enough to separate music from non-music for downstream use.

Primary sources used for this summary

- `saves/lab1_run_combo_af_gate_exit_v2/audits_confidence/leakage_summary.json`
- `saves/lab1_run_combo_af_gate_exit_v2/audits_confidence/gate_summary.json`
- `docs/explanation/results.md`